

Extremizers for the independent Berry–Esseen inequality

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Abstract

Suppose the optimal Berry–Esseen constant for arbitrary independent summands exceeds Esseen’s constant. We prove that the supremum is attained in a class of countable independent convolutions with an infinitely divisible remainder, keeping the sum of third absolute moments as part of the representation. A maximizing representation can be chosen with three-point summands. Its Gaussian and Lévy components vanish, and the supports of its independent summands shrink to zero. If infinitely many summands are nonconstant, their sum is nonatomic. We also prove a rank condition on the finite support configurations that can complete to a maximizing threshold. This condition excludes representations by infinitely many separated finite blocks, without restrictions on their probabilities or cardinalities. The proofs use exact variations of the original laws; CDF discontinuities are retained throughout. The sharp inequality for all independent arrays remains open.

1 Statements and notation

For a centered random variable S of variance one, set

$$\Delta(S) = \sup_{x \in \mathbb{R}} |\mathbb{P}(S \leq x) - \Phi(x)|.$$

The standard normal density and distribution function are denoted by ϕ and Φ . Define

$$C_{\text{ind}} = \sup \frac{\Delta(\sum_{i=1}^n X_i)}{\sum_{i=1}^n \mathbb{E}|X_i|^3}, \quad C_E = \frac{3 + \sqrt{10}}{6\sqrt{2\pi}},$$

where the supremum is over finite independent centered arrays with total variance one and finite third absolute moments. The classical Berry–Esseen theorem gives $C_{\text{ind}} < \infty$, and Esseen’s two-point asymptotics give $C_{\text{ind}} \geq C_E$ [4, 11].

We use the small-Lyapunov theorem proved in the companion manuscript [2, Theorem 1.1]: there is $\ell_0 > 0$ such that

$$\sum_i \mathbb{E}|X_i|^3 \leq \ell_0 \implies \Delta(\sum_i X_i) \leq C_E \sum_i \mathbb{E}|X_i|^3. \quad (1.1)$$

Its threshold is not numerical. That theorem adapts the uniform smoothing, contamination and local comparison methods of He and Cheng [7] to nonidentically distributed arrays. Its finite input is the common-diameter Bernoulli theorem of [1]. The present paper concerns the remaining extremal problem at positive Lyapunov ratio.

Definition 1.1. An admissible representation consists of independent centered variables $X_1, X_2, \dots$, an independent centered infinitely divisible variable Y , and a finite number

$$S = \sum_{i \geq 1} X_i + Y,$$

$$\log \mathbb{E} e^{itY} = -\frac{at^2}{2} + \int_{\mathbb{R} \setminus \{0\}} (e^{ity} - 1 - ity) \nu(dy), \quad (1.2)$$

$$\sum_i \text{Var}(X_i) + a + \int y^2 \nu(dy) = 1, \quad \beta = \sum_i \mathbb{E}|X_i|^3 + \int |y|^3 \nu(dy) < \infty. \quad (1.3)$$

Here $a \geq 0$, the series converges in L^2 , and ν is a Lévy measure with finite second moment. The number β is the additive third-moment quantity of the representation. A representation is maximizing if $\beta > 0$ and $\Delta(S) = C_{\text{ind}}\beta$.

The Gaussian term contributes zero to β . Indeed a Gaussian variable of variance a can be approximated by sums whose variance is a and whose third-moment sums tend to zero. Thus β is not the third absolute moment of S , and it is not determined by the law of S alone. The approximation statement for the whole representation is proved in Section 2.

Theorem 1.2. *If $C_{\text{ind}} > C_E$, there is a maximizing admissible representation whose variables X_i have at most three support points. Every maximizing representation with this support property satisfies*

$$a = 0, \quad \nu = 0, \quad \max\{|u| : u \in \text{supp } X_i\} \rightarrow 0. \quad (1.4)$$

If infinitely many X_i are nonconstant, then the law of S is nonatomic and

$$\sum_i (1 - \max_u \mathbb{P}(X_i = u)) = \infty.$$

Reflection, if needed, gives a threshold z at which

$$F(z) - \Phi(z) = D = C\beta > 0, \quad C = C_{\text{ind}}, \quad A = \frac{3D + z\phi(z)}{2}, \quad F(x) = \mathbb{P}(S \leq x). \quad (1.5)$$

All CDFs in this paper are closed CDFs. A left limit is written explicitly. The function $F - \Phi$ is upper semicontinuous and tends to zero at both ends of the line, so a positive supremum on this branch is attained. The continuous normal CDF allows reflection to interchange the positive branch with the branch $\Phi(x) - \mathbb{P}(S < x)$.

There is also a restriction on support configurations. For a finite set I of indices, put $T_I = S - \sum_{i \in I} X_i$ and

$$\mathcal{A}_I = \left\{ (x_i)_{i \in I} \in \prod_{i \in I} \text{supp } X_i : z - \sum_{i \in I} x_i \in \text{supp } T_I \right\}. \quad (1.6)$$

The support on the right is the closed support of the actual independent remainder. This definition does not condition on the event $S = z$.

Theorem 1.3. *Fix a maximizing representation as in Theorem 1.2 and a threshold satisfying (1.5). There is k_0 such that, for every finite $I \subset \{k_0, k_0 + 1, \dots\}$, the following holds: if functions $f_i : \text{supp } X_i \rightarrow \mathbb{R}$ satisfy*

$$\sum_{i \in I} f_i(x_i) = c \quad \text{for all } (x_i) \in \mathcal{A}_I,$$

then every f_i is constant. Equivalently, choose one state $u_i^0 \in \text{supp } X_i$ for each i . The indicator vectors $(\mathbf{1}_{\{x_i = u\}})_{i \in I, u \in \text{supp } X_i \setminus \{u_i^0\}}$ associated with $\mathcal{A}_I$ have full affine dimension $\sum_{i \in I} (|\text{supp } X_i| - 1)$.

Corollary 1.4. *Such a representation cannot consist of a finite-support independent prefix followed by infinitely many nonconstant finite blocks of finite-support summands for which*

$$H_k = \sum_{i \text{ in block } k} \text{diam}(\text{supp } X_i), \quad \delta_k = \min_{u \neq v \in \text{supp}(\sum_{i \text{ in block } k} X_i)} |u - v|$$

satisfy

$$\sum_k H_k < \infty, \quad \delta_k \geq \sum_{l > k} H_l. \quad (1.7)$$

There are no restrictions on the block sizes, their positive atom masses, or the overlap of the prefix translates.

Theorem 1.2 leaves both finite representations and nonatomic infinite convolutions with shrinking supports. Such infinite convolutions need not have a density. Theorem 1.3 is automatic when every relevant remainder has support $\mathbb{R}$; it therefore does not supply the missing regularity assertion.

The compactification uses the classical accompanying Poisson-law method and Lévy–Khintchine representation [10]; Section 2 proves the needed moment bounds. Coordinate contamination follows He and Cheng [7, Section 4]; the independent-coordinate derivatives, Lévy-intensity variations and contact propagation are derived below. The Gaussian estimates use the zero-bias transformation of Goldstein and Reinert [6], the transport estimate independently obtained by Goldstein and Tyurin [5, 13], and the compound-Poisson Berry–Esseen bound of Korolev and Shevtsova [9]. The atom argument uses Sperner’s theorem [12]. These inputs are stated where they enter the proofs.

2 Compactification and attainment

Lemma 2.1. *For every $c > 0$, a finite independent variance-one array satisfying $\Delta(\sum_i X_i) > c \sum_i \mathbb{E}|X_i|^3$ can be replaced by one with the same number and variances of summands, each having at most three support points, for which the same strict inequality holds.*

Proof. Choose the positive closed-CDF branch by reflection. At its fixed threshold, with all other summands held fixed, maximize

$$\int \{G_i(z - u) - c|u|^3\} P(du) \quad \text{subject to} \quad \int u dP = 0, \quad \int u^2 dP = v_i.$$

For $v_i = 0$ the law is fixed. Otherwise the objective is upper semicontinuous and bounded above by $1 - c|u|^3$. A maximizing sequence has bounded third moments, so a weak limit preserves its first two moments and attains the supremum. The maximizing set is compact and is a face of the convex moment set. It has an extreme point. Such a point has at most three atoms: four disjoint positive-mass sets would give a linear dependence among their mass, first-moment and second-moment vectors, and hence two distinct feasible bounded-density perturbations with the original law as their average. This is the elementary three-constraint case of the moment-set principle in Winkler [14]; the coercive argument is also given in Bell [1]. Replacing coordinates successively preserves the strict positive excess at coefficient c . $\square$

Lemma 2.2. *Let finite independent centered arrays have total variance one and third-moment sums $\beta_n \leq M$. After passage to a subsequence, their sums converge weakly to an admissible representation with*

$$\beta \leq \liminf_n \beta_n.$$

If the original summands have at most three atoms, so do the limiting independent variables X_i .

Proof. First take a subsequence along which β_n tends to its liminf. Order each row by decreasing variance, padding it with zero variables. Then $v_{n,k} \leq 1/k$. The laws are tight, and

$$\mathbb{E}[X_{n,k}^2; |X_{n,k}| > L] \leq M/L$$

makes their squares uniformly integrable. A diagonal subsequence gives $X_{n,k} \Rightarrow X_k$, convergence of means and variances, and convergence of the numbers $\mathbb{E}|X_{n,k}|^3$ to numbers b_k . Writing $v_k = \mathbb{E}X_k^2$ and $r_k = \mathbb{E}|X_k|^3$, we have

$$\sum_k v_k \leq 1, \quad r_k \leq b_k, \quad \sum_k b_k \leq \lim_n \beta_n.$$

The independent series $\sum_k X_k$ exists in L^2 .

Choose $K_n \rightarrow \infty$ sufficiently slowly that the first K_n laws approximate their limits on each fixed prefix and $\sum_{k \leq K_n} |v_{n,k} - v_k| \rightarrow 0$. The corresponding prefix sums converge weakly to $\sum_k X_k$: after fixing a finite prefix, the variance of what remains in the growing prefix is at most $\sum_{k > K} v_k + o(1)$. Define the quadratic measures of the remaining row by

$$\Lambda_n(E) = \sum_{k > K_n} \mathbb{E}[X_{n,k}^2 \mathbf{1}_{\{X_{n,k} \in E\}}].$$

Their masses tend to $v_0 = 1 - \sum_k v_k$ and their tails are bounded by M/L outside $[-L, L]$. Take $\Lambda_n \Rightarrow \Lambda$; its mass is v_0 .

For a centered variable of variance v , its characteristic function f satisfies $|f(t) - 1| \leq vt^2/2$. If $\Re w \leq 0$, then $|e^w - 1 - w| \leq |w|^2/2$. Telescoping products therefore gives

$$\left| \prod_j f_j(t) - \exp\left\{\sum_j (f_j(t) - 1)\right\} \right| \leq \frac{t^4}{8} \sum_j v_j^2. \quad (2.1)$$

For the remaining row the right side tends to zero, since its largest variance is at most $1/(K_n + 1)$. Its accompanying Poisson exponent is

$$\int q_t(y) \Lambda_n(dy), \quad q_t(y) = \frac{e^{ity} - 1 - ity}{y^2}, \quad q_t(0) = -t^2/2.$$

The function q_t is bounded and continuous. Hence these exponents converge to the integral against Λ . Set $a = \Lambda(\{0\})$ and $\nu(dy) = y^{-2}\Lambda(dy)$ off zero. This gives (1.2), with variance v_0 . Independence of the original prefix and tail proves the claimed representation and preserves total variance.

For each fixed K and all sufficiently large n ,

$$\int |y| \Lambda_n(dy) \leq \beta_n - \sum_{k \leq K} \mathbb{E}|X_{n,k}|^3.$$

Lower semicontinuity and then $K \rightarrow \infty$ give $\int |y|^3 \nu(dy) \leq \lim \beta_n - \sum_k b_k$. Together with $r_k \leq b_k$ this proves the third-moment inequality. No uniform integrability of cubic moments was assumed. Finally, a weak limit of three-atom laws cannot charge four disjoint open sets. $\square$

Lemma 2.3. *Every admissible representation is a weak limit of finite centered independent arrays with variance exactly one and third-moment sums tending to its number β . Consequently*

$$\Delta(S) \leq C_{\text{ind}} \beta. \quad (2.2)$$

Proof. Keep finitely many X_i . Truncate ν away from zero and infinity, and approximate it by a finite atomic measure with convergent second and third absolute integrals. A small contraction can make its variance no larger than the available variance. Replace each atom $\lambda\delta_y$ by N independent variables $y(B - \lambda/N)$, where $B \sim \text{Bern}(\lambda/N)$. Their total variance and third-moment sum are

$$\lambda y^2(1 - \lambda/N), \quad \lambda|y|^3(1 - \lambda/N)\{(\lambda/N)^2 + (1 - \lambda/N)^2\}.$$

They converge to the corresponding compensated Poisson law and its second and third jump integrals. Fill the remaining variance by N independent symmetric two-point variables. If that variance is b , their third-moment sum is $b^{3/2}/\sqrt{N}$. Sending the truncation and approximation indices to their limits gives the required Gaussian variance and discards a tail of $\sum_i X_i$ whose variance tends to zero. Every finite array has variance one.

If $P_n \Rightarrow P$ and H is a continuous CDF, then $\|F_{P_n} - H\|_\infty \rightarrow \|F_P - H\|_\infty$. To verify this assertion, sandwich CDF values between a finite grid of continuity points of F_P on which the oscillation of H is arbitrarily small, and control both tails. Values immediately before and after jumps are recovered by continuity points and the continuity of H . Applying this to $H = \Phi$ proves (2.2). $\square$

Proposition 2.4. *If $C_{\text{ind}} > C_E$, a maximizing representation with three-point independent variables exists.*

Proof. Take coefficients $c_n \uparrow C_{\text{ind}}$ with $c_n > C_E$ and finite arrays whose ratios exceed c_n . Lemma 2.1 supplies three-point arrays with the same property. Their third-moment sums satisfy $\ell_0 < \beta_n < 1/c_n$ by (1.1) and $\Delta \leq 1$. Take $\beta_n \rightarrow r > 0$ and apply Lemma 2.2. Weak continuity of distance to the normal gives $\Delta(S) = C_{\text{ind}}r > 0$. The limiting representation satisfies

$$C_{\text{ind}}r = \Delta(S) \leq C_{\text{ind}}\beta \leq C_{\text{ind}}r.$$

Thus $\beta = r$ and it is maximizing. This equality rules out loss of the additive third-moment quantity in the compactification. $\square$

3 Variations and contact identities

Fix a maximizing representation with $C = C_{\text{ind}} > C_E$ and (1.5). For an admissible perturbation of raw independent laws, let m be the total mean, B the total central variance, R_* the sum of the recentered third absolute moments plus the Lévy third integral, and F_* the raw sum CDF. At a raw threshold w , put

$$J = B^{3/2} \left\{ F_*(w) - \Phi\left(\frac{w - m}{\sqrt{B}}\right) \right\} - CR_*. \quad (3.1)$$

By Lemma 2.3, $J \leq 0$ whenever the perturbation is admissible, and $J = 0$ at the original representation and $w = z$. Finite probability variations make $F_*(z)$ a multiaffine polynomial, including at atoms. Since $|x|^3$ is C^2 , their central third moments are twice differentiable. A two-sided probability parameter consequently has zero first derivative, and the Hessian is negative semidefinite on its admissible tangent cone when all first derivatives vanish.

Lemma 3.1. *Let G_i be the CDF of $S - X_i$, and put $v_i = \mathbb{E}X_i^2$, $b_i = \mathbb{E}|X_i|^3$, $M_i = \mathbb{E}[X_i|X_i|]$. Then*

$$G_i(z - u) - F(z) + \phi(z)u + A(u^2 - v_i) - C\{|u|^3 - b_i - 3M_i u\} \leq 0, \quad (3.2)$$

$$Q(y) := F(z - y) - F(z) + \phi(z)y + Ay^2 - C|y|^3 \leq 0. \quad (3.3)$$

Equality holds in (3.2) at every support point of X_i , and $Q(y) = 0$ at every nonzero point of $\text{supp } \nu$. All supports $\text{supp } X_i$ lie in one bounded interval, and all nonzero zeros of Q lie in one bounded interval.

Proof. Replace the raw law of X_i by $(1-t)\text{Law}(X_i) + t\delta_u$. The derivatives of mean, variance and third absolute central moment are u , $u^2 - v_i$ and $|u|^3 - b_i - 3M_i u$. Differentiating (3.1) gives (3.2). Its integral under $\text{Law}(X_i)$ is zero. Upper semicontinuity then extends equality from almost every point to the entire support. This is the coordinatewise contamination argument of He and Cheng [7, Section 4].

Append a compensated Poisson variable $y(N_t - t)$ and use the centered threshold $z - ty$. Its raw CDF is exactly $\mathbb{E}F(z - yN_t)$; variance and third-moment quantity increase by ty^2 and $t|y|^3$. Its right first derivative is $Q(y)$, proving (3.3). For a finite signed measure $\eta = f\nu$, with bounded f supported away from zero, both signs of small changes in ν are admissible. Their derivatives are $\int Q d\eta$. Thus $Q = 0$ almost everywhere for ν , and upper semicontinuity gives equality on its nonzero support.

In (3.2), use $v_i \leq 1$, $|M_i| \leq 1$ and $Cb_i \leq C\beta = D$. Its right side at a support point is bounded above by a fixed cubic polynomial in $|u|$ with leading coefficient $-C$. The same reasoning applies to (3.3). Both zero sets are therefore bounded uniformly in i . $\square$

For later calculations, consider a mass transfer between two states u, v of one centered variable. Write $d = u - v$, $b = u^2 - v^2$, and let $g = G_i(z - u) - G_i(z - v)$. The third-moment derivatives are

$$\begin{aligned} r' &= |u|^3 - |v|^3 - 3d\mathbb{E}[X_i|X_i|], \\ r'' &= -6d(u|u| - v|v|) + 6d^2\mathbb{E}|X_i|. \end{aligned} \quad (3.4)$$

Stationarity says $g + \phi d + Ab - Cr' = 0$. Hereafter ϕ without an argument in an algebraic formula means $\phi(z)$. Put

$$B_0 = \frac{3D + (3z + z^3)\phi}{4}, \quad E_0 = (1 + z^2/2)\phi.$$

For two transfers in distinct coordinates the Hessian entries of (3.1) are

$$\begin{aligned} H_{ii} &= 3b_i g_i + B_0 b_i^2 + 2E_0 b_i d_i - 3D d_i^2 - Cr''_i, \\ H_{ij} &= g_{ij} + \frac{3}{2}(b_i g_j + b_j g_i) + B_0 b_i b_j + E_0(b_i d_j + b_j d_i) + z\phi d_i d_j, \quad i \neq j. \end{aligned} \quad (3.5)$$

Here g_{ij} is the exact raw double CDF difference. These identities follow by differentiating $B^{3/2}\Phi((z - m)/\sqrt{B})$ at $(B, m) = (1, 0)$; its second derivatives are

$$N_{mm} = -z\phi, \quad N_{mB} = -(1 + z^2/2)\phi, \quad N_{BB} = \frac{3}{4}\Phi(z) - \frac{1}{4}(3z + z^3)\phi.$$

The second variance derivative in one transfer is $-2d_i^2$, whereas mixed moment derivatives in different coordinates vanish. This accounts for the diagonal term $-3Dd_i^2$ in (3.5).

Lemma 3.2. *If the full sum has an atom of mass $m > 0$ and each X_i has at most three support points, then, with $q_i = \max_u \mathbb{P}(X_i = u)$,*

$$\sum_i (1 - q_i) < \infty.$$

Proof. Represent each X_i as an activated fair two-point law. If $q_i \geq 1/2$, activate with probability $\rho_i = 2(1 - q_i)$ and use the modal state with the remaining probability. On activation choose fairly between the modal state and an independent conditional nonmodal state. If three masses p_1, p_2, p_3 are all below $1/2$, mix the three fair pairs, assigning weight $1 - 2p_k$ to the pair that excludes state k . Thus in all cases $\rho_i = \min\{1, 2(1 - q_i)\} \geq \frac{3}{2}(1 - q_i)$.

Condition on activations and chosen endpoints, orienting the fair bits with positive increments. Every fixed-sum fiber is an antichain. Sperner's theorem [12] bounds its conditional probability

by $2^{-N} \binom{N}{\lfloor N/2 \rfloor} \leq \sqrt{2/(N+1)}$ when N bits are active. For a finite prefix with $\mu = \sum_i \rho_i > 0$, Cauchy–Schwarz and the generating function of N yield

$$\sup_x \mathbb{P}(\sum_i X_i = x)^2 \leq 2\mathbb{E} \frac{1}{N+1} = 2 \int_0^1 \prod_i (1 - \rho_i + \rho_i t) dt \leq \frac{2(1-e^{-\mu})}{\mu}.$$

Convolution cannot increase the largest atom. The left side is at least m^2 for every prefix, so μ stays bounded, proving the assertion. This is a direct finite-alphabet concentration argument within the classical theory of infinite convolutions [8]. $\square$

4 Eliminating jumps and nonshrinking supports

4.1 Pair contacts for the Lévy measure

Lemma 4.1. *For nonzero same-sign points $x, y \in \text{supp } \nu$, one has $Q(x+y) = 0$. The measure ν is supported on at most two positive and two negative points.*

Proof. Choose a probability law H dominated by a finite multiple of ν and supported in a compact interval on one side of zero. Write $m = \mathbb{E}H$, $v = \mathbb{E}H^2$, $r = \mathbb{E}|H|^3$, and $k = m^2/v$. Remove a compensated Poisson component of intensity $p - kp^2$ with jump law H , and insert the single centered variable $B_p H - pm$. This is admissible for all sufficiently small $p > 0$ and preserves variance. Its third absolute moment is

$$pr - 3p^2|m|v + 4p^3|m|^3 - 2p^4|m|^3.$$

At the centered threshold $z - kp^2m$, the raw CDF is

$$[(I + pU_H)e^{-(p-kp^2)U_H}F](z), \quad U_H f(w) = \mathbb{E}f(w-H) - f(w).$$

The bounded-operator expansion is valid in uniform norm. Its second-order term is $(kU_H - \frac{1}{2}U_H^2)F$. Combining it with the normal and third-moment terms, the coefficient of p^2 in the excess is

$$(k+1)\mathbb{E}Q(H) - \frac{1}{2}\mathbb{E}Q(H+H'),$$

where H' is independent. The cubic cancellation uses $\mathbb{E}|H+H'|^3 = 2r + 6|m|v$. The first expectation vanishes by Lemma 3.1; the second is nonpositive. Maximality forces it to be zero. Upper semicontinuity extends this identity to all same-sign support pairs, by choosing H on neighborhoods of any two specified support points. The case $x = y$ is included.

For the support count, perturb the Lévy measure by a finite signed measure $\eta = f\nu$ with bounded f supported away from zero. Define $d_\eta = \int y d\eta$, $v_\eta = \int y^2 d\eta$, and $r_\eta = \int |y|^3 d\eta$. The centered threshold $z - d_\eta$ gives

$$(1 + v_\eta)^{3/2} \left\{ [e^{U_\eta} F](z) - \Phi \left(\frac{z - d_\eta}{\sqrt{1 + v_\eta}} \right) \right\} - C(\beta + r_\eta) \leq 0, \quad (4.1)$$

where $U_\eta f(w) = \int [f(w-y) - f(w)] \eta(dy)$. Both signs are allowed near zero. Negative perturbations are interpreted by removing the corresponding finite Poisson component; the bounded operator identity remains exact. Thus its Hessian is negative semidefinite on these signed directions.

Write $U_x f(w) = f(w-x) - f(w)$. Let $L = (15D + \phi'''(z))/4$, and, for $\sigma \in \{-1, 1\}$, define

$$\mathcal{K}_\sigma(u, v) = uv \left\{ -3D + \left(3C + \frac{\sigma \phi''(z)}{2} \right) (u+v) - Luv + \frac{3C}{2} uv(u+v) \right\}, \quad u, v > 0. \quad (4.2)$$

At same-sign contact points $x = \sigma u$, $y = \sigma v$ whose sum is also a contact, the intensity Hessian equals $\mathcal{K}_\sigma(u, v)$. For verification before substituting contacts, its kernel is

$$U_x U_y F(z) + \frac{3}{2}[x^2 U_y F(z) + y^2 U_x F(z)] + B_0 x^2 y^2 + E_0(x^2 y + x y^2) + z \phi x y.$$

Substitute $U_x F(z) = -\phi x - Ax^2 + C|x|^3$ and $U_x U_y F(z) = -2Axy + 3Cxy(|x| + |y|)$ to obtain (4.2). Its moment coefficient matrix in the basis (u, u^2, u^3) is

$$M_\sigma = \begin{pmatrix} -3D & 3C + \sigma\phi''(z)/2 & 0 \\ 3C + \sigma\phi''(z)/2 & -L & 3C/2 \\ 0 & 3C/2 & 0 \end{pmatrix}. \quad (4.3)$$

It is not negative semidefinite: its value at $(0, t, 1)$ is $-Lt^2 + 3Ct > 0$ for small $t > 0$.

If three distinct support magnitudes u_1, u_2, u_3 existed on one side, restrict ν to three disjoint small compact neighborhoods and normalize each restriction to a probability measure. Arbitrary signed combinations are locally admissible intensity directions. Their Hessian is congruent to M_σ through the columns of first, second and third moments of these restrictions. As the neighborhoods shrink, these columns tend to (u_j, u_j^2, u_j^3) , an invertible Vandermonde matrix. For sufficiently small neighborhoods the Hessian cannot be negative semidefinite. This contradiction proves the support count. Finite second moment then makes ν a finite measure. $\square$

4.2 Contacts from limiting support values

Lemma 4.2. *Suppose i_n are distinct and $u_n \in \text{supp } X_{i_n}$ satisfies $u_n \rightarrow x \neq 0$. Then $Q(x) = Q(2x) = 0$. The probability transfer can be taken from a state $v_n \rightarrow 0$ to u_n , with gap $d_n = u_n - v_n \rightarrow x$. If the full CDF has a jump and $d_n > x$ along a subsequence, it has no jump at $z - x$ or $z - 2x$.*

Proof. The variances of distinct coordinates tend to zero. If F is continuous, choose a support state v_n with $|v_n| \leq \sqrt{v_{i_n}}$. Deleting one or two selected coordinates changes the sum by a variable tending to zero in L^2 . The deleted CDFs therefore converge uniformly to F . The derivatives in (3.4) satisfy

$$d_n \rightarrow x, \quad b_n = u_n^2 - v_n^2 \rightarrow x^2, \quad r'_n \rightarrow |x|^3, \quad r''_n \rightarrow -6|x|^3.$$

First stationarity gives $Q(x) = 0$. In (3.5), the limit of the off-diagonal entry minus that of the diagonal entry is

$$F(z - 2x) - 2F(z - x) + F(z) + 2Ax^2 - 6C|x|^3 = Q(2x).$$

The two-coordinate contrast $(1, -1)$ gives $-2Q(2x) \leq 0$. Together with $Q \leq 0$, this proves the assertion.

If the full law has an atom, Lemma 3.2 gives modal probabilities $q_n \rightarrow 1$. Take v_n to be a modal state; then $q_n v_n^2 \leq v_{i_n}$, so $v_n \rightarrow 0$. Put $H_n(w) = \mathbb{P}(S - X_{i_n} + v_n \leq w)$. Since the original law contains this branch with probability q_n ,

$$\|H_n - F\|_\infty \leq 1 - q_n.$$

The analogous double-deletion bound is $(1 - q_n) + (1 - q_m)$. Pass to a subsequence on which the gaps equal x , lie above x , or lie below x . Each shifted CDF then has a definite side limit. Let Q_1^*, Q_2^* denote the corresponding versions of $Q(x), Q(2x)$, using these side limits. Stationarity gives $Q_1^* = 0$; the same Hessian contrast gives $Q_2^* \geq 0$, while $Q \leq 0$ gives $Q_2^* \leq 0$. Closed CDF values dominate their left limits, so $Q(x), Q(2x) \geq 0$. Both are zero. When the arguments approach from the left, the two side limits must equal the closed values; this proves the last assertion. $\square$

4.3 Comparisons with fixed added summands

Fix a sign σ and a nonzero limit $x = \sigma u$ as in Lemma 4.2. Its probability transfers are two-sided at each finite index. Appending a fresh raw variable $(1 - t)\delta_0 + t\delta_y$, $t \geq 0$, has first derivative $Q(y)$ in (3.1). The gap y below is always fixed, rather than set equal to a varying support value. Thus when $Q(y) = 0$, its first derivative is exactly zero for every selected index.

Lemma 4.3. *For $y = \sigma v$ with $Q(y) = 0$, the limiting mixed Hessian between the old transfer tending to x and a fresh variable at y equals*

$$\mathcal{K}_\sigma(u, v) + Q(x + y).$$

The old diagonal and its mixed entry with a fresh variable at x both equal $\mathcal{K}_\sigma(u, u)$ in the limit. The same conclusions hold if the old direction instead varies the positive intensity of an existing Lévy atom at x .

Proof. The limiting derivatives of an old probability transfer are

$$m' = x, \quad B' = x^2, \quad B'' = -2x^2, \quad R'_* = |x|^3, \quad R''_* = -6|x|^3.$$

They coincide with the derivatives for the fresh variable at x . Stationarity and (3.5) give the common limiting diagonal $\mathcal{K}_\sigma(u, u)$. Their mixed CDF double shift is closed: in the continuous case this follows from uniform deletion convergence; in the atomic case its argument approaches $z - 2x$ from the right, or Lemma 4.2 removes the left-limit jump. Since $Q(x) = Q(2x) = 0$, its mixed entry is the same diagonal.

Append also a fresh variable at the fixed contact y . Pass to a subsequence of the bounded Hessian entries. Write B_{xy} for the old–fresh- y entry and N_{xy} for the fresh- x –fresh- y entry. The latter uses the exact closed value $F(z - x - y)$ and, by the same substitution as in (4.2), equals $\mathcal{K}_\sigma(u, v) + Q(x + y)$. The old single-shift limit is fixed by stationarity; the fresh single shift is closed. Upper semicontinuity of the double shift, together with the uniform deletion or modal-branch comparison in the preceding lemma, therefore gives $N_{xy} - B_{xy} \geq 0$.

The vector $w = (-1, 1, 0)$ has zero limiting quadratic value and is feasible: it decreases an old probability and increases its fresh copy. For every $t \geq 0$, $w + t(0, 0, 1)$ is also a feasible tangent direction. The Hessian inequality gives

$$2t(N_{xy} - B_{xy}) + t^2 H_{yy} \leq 0.$$

Letting $t \downarrow 0$ proves the reverse inequality. This identifies the closed mixed entry. All first derivatives were exactly zero before taking the limit, so no approximation of a first-order contact is used to infer a second-order inequality.

For an existing Lévy atom, vary its intensity in both directions. Its CDF and moment derivatives are already at fixed closed shifts. Relative to the fresh Bernoulli diagonal its second derivative differs by

$$U_x^2 F(z) + 2Ax^2 - 6C|x|^3 = Q(2x) - 2Q(x) = 0,$$

using Lemma 4.1. This proves the same formulas. □

Proposition 4.4. *Every maximizing representation with three-point independent variables has $\nu = 0$ and $\max\{|u| : u \in \text{supp } X_i\} \rightarrow 0$.*

Proof. Let $x = \sigma u$ be either a nonzero limiting support value or a point of $\text{supp } \nu$. In the latter case it is an atom of positive intensity by Lemma 4.1. Let $y = \sigma v$ be any same-sign global contact. For fixed $\varepsilon > 0$, append the one raw three-point law

$$(1 - (1 + \varepsilon)t)\delta_0 + t\delta_x + \varepsilon t\delta_y, \quad t \geq 0.$$

Its first derivative is $Q(x) + \varepsilon Q(y) = 0$. If $m_1 = x + \varepsilon y$, $v_1 = x^2 + \varepsilon y^2$, and $s_1 = x|x| + \varepsilon y|y|$, its moment derivatives are

$$m' = m_1, \quad B' = v_1, \quad B'' = -2m_1^2, \quad R'_* = |x|^3 + \varepsilon|y|^3, \quad R''_* = -6m_1s_1.$$

The raw CDF is affine in t . Its Hessian diagonal is consequently

$$\mathcal{K}_\sigma(u, u) + 2\varepsilon\mathcal{K}_\sigma(u, v) + \varepsilon^2\mathcal{K}_\sigma(v, v).$$

By Lemma 4.3, its mixed entry with the old direction is $\mathcal{K}_\sigma(u, u) + \varepsilon[\mathcal{K}_\sigma(u, v) + Q(x + y)]$. Evaluate the two-parameter limiting Hessian on $(-1, 1)$, decreasing the old direction and increasing the new activation. It gives

$$-2\varepsilon Q(x + y) + \varepsilon^2\mathcal{K}_\sigma(v, v) \leq 0.$$

First take the old-coordinate limit with ε fixed, and only then let $\varepsilon \downarrow 0$. Thus $Q(x + y) \geq 0$, and hence $Q(x + y) = 0$. The admissible neighborhoods of the old probabilities need not have a uniform radius.

Starting with $Q(x) = 0$, this conclusion propagates to $Q(nx) = 0$ for every positive integer n , always using the same old direction and a fixed contact at the next step. This contradicts the bounded contact set in Lemma 3.1. No such x exists. The finite Lévy measure must therefore be zero. The uniformly bounded independent supports cannot have a nonzero accumulation along distinct indices, so their radii tend to zero. $\square$

5 Excluding a Gaussian component

By Proposition 4.4, it is enough to consider

$$S = W + G_a, \quad W = \sum_i X_i, \quad \text{Var}(W) = 1 - a, \quad \beta = \sum_i \mathbb{E}|X_i|^3,$$

where $G_a \sim N(0, a)$ is independent. Suppose $a > 0$. The law has a smooth density f . Threshold and Gaussian-variance stationarity give

$$f(z) = \phi(z), \quad f'(z) = -3D - z\phi(z), \quad |f''(z)| \leq 6C. \quad (5.1)$$

The second identity follows from the heat equation and the derivatives of (3.1). Both signs of a small variance change are allowed. The last inequality follows by expanding Q at zero on its two sides.

5.1 A smoothing bound from zero bias

For a centered variable T of variance $v > 0$, its zero-biased law T^* is characterized by $\mathbb{E}[Tg(T)] = v\mathbb{E}g'(T^*)$. Goldstein and Reinert [6] introduced this transformation and the independent-sum replacement construction. Goldstein [5, Lemma 1.1] and Tyurin [13, Theorem 3 and Corollary 1] give

$$W_1(W, W^*) \leq \frac{\beta}{2(1 - a)}, \quad (5.2)$$

where W_1 is Wasserstein distance with cost $|x - y|$. For countably many summands the statements below follow by finite approximation retaining the fixed Gaussian variance.

Lemma 5.1. *For every such independent sum,*

$$\Delta(W + G_a) \leq K(a)\beta, \quad K(a) = \frac{\phi(0)}{2} \int_0^1 \frac{s^2}{[1 - (1 - a)s^2]^{3/2}} ds.$$

The function K is decreasing, and $K(1/10) < C_E$. Hence a maximizing representation has $a < 1/10$.

Proof. For $h(u) = \Phi((z - u)/\sqrt{a})$ and $v = 1 - a$, the solution of $vg'(u) - ug(u) = h(u) - \mathbb{E}h(\sqrt{v}Z)$ has the Ornstein–Uhlenbeck representation

$$g(u) = - \int_0^1 \mathbb{E}h'(su + \sqrt{v(1 - s^2)}Z) ds.$$

Differentiating twice and convolving Gaussian derivatives gives $\|g''\|_\infty \leq \phi(0) \int_0^1 s^2(1 - vs^2)^{-3/2} ds$. The zero-bias identity and (5.2) bound the expectation difference by $\beta\|g''\|_\infty/2$. The integral decreases with a , and direct integration gives

$$K(a) = \frac{\phi(0)}{2} \left\{ \frac{1}{(1 - a)\sqrt{a}} - \frac{\arcsin \sqrt{1 - a}}{(1 - a)^{3/2}} \right\}.$$

At $a = 1/10$ this is $(5\sqrt{10}/27)\phi(0)(3 - \arctan 3)$. To check the strict comparison without decimal rounding, use $\pi > 157/50$, $\sqrt{10} < 19/6$ and

$$\arctan 3 = \pi/2 - \arctan(1/3) > \frac{157}{100} - \left(\frac{1}{3} - \frac{1}{81} + \frac{1}{1215} \right) > \frac{237}{190} > \frac{21}{10} - \frac{27}{10\sqrt{10}}.$$

This is equivalent to $K(1/10) < C_E$. □

5.2 A lattice comparison

Lemma 5.2. *At a maximizing representation with $a > 0$,*

$$\phi(z) \leq a \left(C - \frac{f''(z)^2}{36C} \right). \quad (5.3)$$

Proof. We first approximate $W + G_a$ by a lattice convolution plus a Gaussian, with error smaller than its lattice spacing. Order the X_i by decreasing variance and set

$$v^{(K)} = \sum_{i>K} v_i, \quad H_K = \sum_{i>K} v_i^2.$$

Then $KH_K \leq v^{(K)} \rightarrow 0$. If the series is infinite, choose $h_K = \sqrt{H_K/K}$, so $H_K = o(h_K)$ and $Kh_K \rightarrow 0$. For finitely many nonzero summands take $h_K = K^{-2}$ beyond them.

Replace the tail after K by its accompanying compensated Poisson law with measure $\mu_K = \sum_{i>K} \text{Law}(X_i)|_{\mathbb{R} \setminus \{0\}}$. Variance and additive third moment are unchanged. By (2.1), after convolution with G_a the uniform CDF error is at most

$$\frac{H_K}{8\pi} \int_0^\infty t^3 e^{-at^2/2} dt = \frac{H_K}{4\pi a^2} = o(h_K).$$

Extra powers of t give the corresponding density-derivative estimates.

Round each of the first K summands to its neighboring points on $h_K\mathbb{Z}$, independently and with its conditional mean unchanged. Writing the rounding error as δ , we have $\mathbb{E}(\delta \mid X) = 0$, $|\delta| \leq h_K$ and $\mathbb{E}(\delta^2 \mid X) \leq h_K^2/4$. Subtract its total added variance from the Gaussian component. Gaussian Taylor estimates bound the CDF change by $O_a(Kh_K^2) = o(h_K)$ and likewise control fixed density derivatives. The third-moment change is at most

$$3h_K^2 \sum_{i \leq K} \mathbb{E}|X_i| + Kh_K^3 = o(h_K),$$

since the sum of first absolute moments is at most $\sqrt{K}$.

Replace the jumps of μ_K smaller than h_K in magnitude by their Gaussian variance, and round all larger jumps without changing their conditional means. Subtract their added variance from the Gaussian part. For a large jump y , with rounding variance $e_h(y)$, the compensated exponent error is bounded by

$$\left| \mathbb{E}e^{it(y+\delta)} - e^{ity} + \frac{1}{2}t^2 e_h(y) \right| \leq |t|^3 \left(\frac{1}{2}h^2|y| + \frac{1}{6}h^3 \right).$$

For a small jump it is at most $|t|^3|y|^3/6$. Integration against μ_K gives exponent error $O(|t|^3 h_K v^{(K)})$ and third-moment change $O(h_K v^{(K)}) = o(h_K)$. The Gaussian variance changes by $o(1)$ and remains at least $a/2$. Factoring a common smaller Gaussian factor, the bound $|e^u - e^v| \leq |u - v|$ for $\Re u, \Re v \leq 0$ justifies the CDF and derivative comparisons by Fourier inversion.

We have obtained variance-one representations $T_j + G_{a_j}$, where T_j lies on a translate of $h_j\mathbb{Z}$, $a_j \rightarrow a$, and

$$\beta_j = \beta + o(h_j), \quad \|F_{T_j+G_{a_j}} - F\|_\infty = o(h_j), \quad f_j^{(k)} \rightarrow f^{(k)} \text{ uniformly}, \quad k = 0, 1, 2. \quad (5.4)$$

The rounded jump measure has finite intensity, because the pre-rounded jumps have magnitude at least h_j and finite second integral. Its centering is well-defined. All approximations are admissible representations.

Fix $p \in (0, 1)$. Choose integers n_j and $p_j \rightarrow p$ so that $n_j h_j^2 p_j (1 - p_j) = a_j$; rounding $a_j/[h_j^2 p(1 - p)]$ to a nearby integer and solving for p_j does this, with the upper integer used when $p = 1/2$. Put $u_j = 1 - 2p_j$ and $B_j = h_j(\text{Bin}(n_j, p_j) - n_j p_j)$. The lattice Edgeworth expansion at closed endpoints is

$$\mathbb{P}(B_j \leq x) = \Phi_{a_j}(x) + \frac{h_j}{2} \phi_{a_j}(x) - \frac{a_j h_j u_j}{6} \phi_{a_j}''(x) + o(h_j), \quad (5.5)$$

uniformly in those endpoints. Here Φ_b, ϕ_b denote the centered normal CDF and density of variance b . This is the binomial case of the classical lattice expansion of Esseen; the closed-endpoint convention is recorded, for example, in Dinev and Mattner [3, equation (3)]. For the uniformity with $p_j \rightarrow p$, Stirling's formula gives, with $\sigma_j = \sqrt{n_j p_j (1 - p_j)}$ and $t_k = (k - n_j p_j)/\sigma_j$,

$$\mathbb{P}(\text{Bin}(n_j, p_j) = k) = \frac{\phi(t_k)}{\sigma_j} \left[1 + \frac{u_j(t_k^3 - 3t_k)}{6\sigma_j} + O\left(\frac{1 + |t_k|^6}{n_j}\right) \right]$$

uniformly for $|t_k| \leq \sqrt{B \log n_j}$ and p_j in a fixed compact subinterval. Chernoff bounds control the omitted tails by $o(n_j^{-1/2})$ when B is sufficiently large. Summation of the remainder is $O((\log n_j)^3/n_j)$, and right-endpoint Euler–Maclaurin summation adds $\phi(t_k)/(2\sigma_j)$. Integration of the cubic Hermite term is $(1 - t_k^2)\phi(t_k)$, proving (5.5).

Choose z_j on the full lattice of $T_j + B_j$ with $|z_j - z| \leq h_j/2$. Conditional on every value of T_j , the endpoint $z_j - T_j$ is a closed binomial endpoint, so (5.5) gives

$$F_{T_j+B_j}(z_j) = F_{T_j+G_{a_j}}(z_j) + \frac{h_j}{2} f_j(z_j) - \frac{a_j h_j u_j}{6} f_j''(z_j) + o(h_j).$$

Threshold stationarity and (5.4) make its base excess $D + o(h_j)$. The actual third-moment sum of the binomial carrier is $a_j h_j(1 + u_j^2)/2$. Maximality therefore implies, for $u = 1 - 2p$,

$$\phi(z) - \frac{au}{3} f''(z) - Ca(1 + u^2) \leq 0.$$

The inequality extends to $u \in [-1, 1]$ by continuity. Optimize at $u = -f''(z)/(6C)$, which lies in this interval by (5.1), to obtain (5.3). $\square$

Gaussian convolution gives the semiconvexity inequality $f'(z)^2 \leq f(z)f''(z) + f(z)^2/a$ by Cauchy–Schwarz on its Gaussian mixture formula. Set $c = \phi(z)/(aC)$. Lemma 5.2 gives $0 < c \leq 1$ and $|f''(z)| \leq 6C\sqrt{1-c}$. Hence

$$(3D + z\phi(z))^2 \leq aC^2\{c^2 + 6c\sqrt{1-c}\} \leq \frac{45}{16}aC^2.$$

The last maximum is at $c = 3/4$. Moreover, $|z|\phi(z) \leq \sqrt{2\phi(0)/e}\sqrt{\phi(z)}$. Since $C > 2/5$, $\phi(0) < 2/5$ and $e > 27/10$, it follows that

$$\beta \leq \left(\frac{\sqrt{5}}{4} + \frac{\sqrt{2\phi(0)/e}}{3\sqrt{C}} \right) \sqrt{a} < \frac{17}{20} \sqrt{a}.$$

For the final rational comparison one may use $\sqrt{5}/4 < 14/25$ and $\sqrt{20/27}/3 < 29/100$. Together with Lemma 5.1, this confines a possible Gaussian component to

$$0 < \beta < \frac{269}{1000}, \quad \frac{400}{289}\beta^2 < a < \frac{1}{10}. \quad (5.6)$$

5.3 Fourier exclusion of the remaining window

Korolev and Shevtsova [9, Section 3.3, Theorem 2] prove that a centered compound-Poisson variable of variance $v > 0$ and third absolute jump integral r satisfies

$$d_K(Y, N(0, v)) \leq 0.3051 r/v^{3/2}. \quad (5.7)$$

Their jump law is assumed to have positive variance. A vanishing symmetric perturbation of a constant jump law removes this restriction. The same bound holds for every finite-variance infinitely divisible law with third jump integral r : truncate its Lévy measure and replace the omitted variance and Gaussian part by symmetric jumps of size η . These add at most $v\eta$ to the third jump integral. Pass to the limit using weak continuity of distance to the continuous normal CDF.

Lemma 5.3. *A variance-one independent sum satisfying (5.6) has $\Delta(S) < (2/5)\beta$.*

Proof. The elementary inequality

$$\cos x \leq 1 - x^2/2 + |x|^3/10 \quad (5.8)$$

holds on the real line. For $0 \leq x \leq 12/5$ use the fourth-degree cosine upper Taylor bound, and for $x \geq 5$ its right side is at least one. On $12/5 \leq x \leq 5$, the inequality $\cos x \leq -1 + (x - \pi)^2/2$ reduces the claim to positivity of $P(x) = 2 - x^2 + \pi x - \pi^2/2 + x^3/10$. With $u = x - 4$,

$$P(x) = P(4) + (\pi - 16/5)u + u^2(1/5 + u/10).$$

For $-8/5 \leq u \leq 0$ this is at least $P(4)$, and for $0 \leq u \leq 1$ it is at least $P(4) - (5/4)(\pi - 16/5)^2$. Both are positive from $157/50 < \pi < 22/7$. Evenness proves (5.8).

For independent centered copies X, X' , the pointwise inequality

$$|x - y|^3 \leq 2(|x|^3 + |y|^3) - 2xy(|x| + |y|)$$

gives $\mathbb{E}|X - X'|^3 \leq 4\mathbb{E}|X|^3$. Thus a coordinate characteristic function h_i obeys

$$|h_i(t)| \leq \exp\{-v_i t^2/2 + b_i |t|^3/5\}, \quad b_i = \mathbb{E}|X_i|^3. \quad (5.9)$$

Write h for the full characteristic function. On omitting coordinate i , (5.9) and the exact Gaussian factor give

$$e^{-at^2/2} \prod_{j \neq i} |h_j(t)| \leq e^{25/54 - t^2/2 + \beta t^3/5}, \quad t \geq 0.$$

Indeed $b_i \geq v_i^{3/2}$ and $\max_{u \geq 0}(u^2/2 - u^3/5) = 25/54$. The single-coordinate form of (5.2) gives $|h_i(t) - h_i^*(t)| \leq b_i t/(2v_i)$. Selecting the replaced coordinate with its variance probability in the zero-bias identity yields

$$|h'(t) + th(t)| \leq \frac{\beta}{2} t^2 e^{25/54 - t^2/2 + \beta t^3/5}.$$

Multiply by $e^{t^2/2}$ and integrate from zero. Then

$$|h(t) - e^{-t^2/2}| \leq \frac{5}{6} e^{25/54 - t^2/2} (e^{\beta t^3/5} - 1) \leq \frac{\beta}{6} e^{25/54} t^3 e^{-t^2/2 + \beta t^3/5}. \quad (5.10)$$

These calculations first apply to a finite sum. Finite approximations retaining a and converging in variance and third-moment sum justify both inequalities for the countable sum.

Set $c = 19/10$ and $T = c/\beta > 7$. On $[0, 6]$ use $\beta \leq 269/1000$, and on $[6, T]$ use $\beta t \leq c$. It follows that

$$\frac{1}{\pi\beta} \int_0^T \frac{|h(t) - e^{-t^2/2}|}{t} dt \leq \frac{e^{25/54}}{6\pi} (I + J), \quad (5.11)$$

where

$$I = \int_0^6 t^2 e^{-t^2/2 + (269/5000)t^3} dt, \quad J = \int_6^\infty t^2 e^{-3t^2/25} dt.$$

For the high frequencies, $|h_W(t)|$ is bounded by the nonnegative characteristic function of a symmetric infinitely divisible variable Π with Lévy measure $\frac{1}{2} \sum_i \text{Law}(X_i - X'_i)$ off zero. This follows from $|h_i| \leq \exp[-(1 - |h_i|^2)/2]$. Its variance is $V = 1 - a$ and its third jump integral is at most 2β . By (5.7), $\Delta(\Pi, N(0, V)) \leq 0.6102\beta/V^{3/2}$. Integration by parts against a Gaussian density of variance $a/2$ gives

$$\begin{aligned} f_{\Pi + G_{a/2}}(0) &\leq \frac{\phi(0)}{\sqrt{V + a/2}} + \frac{0.6102\beta}{V^{3/2}} \frac{\sqrt{2/\pi}}{\sqrt{a/2}} \\ &< \frac{\phi(0)}{\sqrt{0.95}} + \frac{0.6102}{0.9^{3/2}} \frac{17}{20} \frac{2}{\sqrt{\pi}} < \frac{11}{10}. \end{aligned}$$

The first inequality uses the total variation of the Gaussian density; the second uses (5.6). The nonnegative characteristic function of Π and exact density inversion now give

$$\frac{1}{\pi} \int_T^\infty \frac{e^{-at^2/2} |h_W(t)|}{t} dt \leq \frac{e^{-aT^2/4}}{T} f_{\Pi+G_{a/2}}(0).$$

Consequently its ratio to β is less than $(11/19)e^{-361/289} < 1/6$. The corresponding normal tail is at most

$$\frac{1}{\pi\beta} \int_T^\infty \frac{e^{-t^2/2}}{t} dt \leq \frac{\beta}{\pi c^2} e^{-c^2/(2\beta^2)} < 10^{-9}.$$

Appendix A supplies the complete rational checks

$$I < 23/10, \quad J < 3/8, \quad e^{25/54}/(6\pi) < 17/200,$$

and the two scalar high-frequency comparisons just used. Exact CDF inversion is applicable because the sum has Gaussian variance $a > 0$. Combining its two frequency ranges proves

$$\frac{\Delta(S)}{\beta} < \frac{17}{200} \left(\frac{23}{10} + \frac{3}{8} \right) + \frac{1}{6} + 10^{-9} = \frac{1182125003}{3000000000} < \frac{2}{5}.$$

□

Equations (5.6) and Lemma 5.3 contradict $C > C_E > 2/5$. This proves $a = 0$ in Theorem 1.2.

6 An infinite maximizing sum is nonatomic

We now have $S = \sum_i X_i$, with uniformly bounded supports whose radii tend to zero. Suppose this full law has an atom. We prove that only finitely many variables can be nonconstant.

Choose a modal state c_i of probability q_i , and set $p_i = 1 - q_i$. Lemma 3.2 gives $\sum_i p_i < \infty$. Centering gives

$$|c_i| \leq \sqrt{v_i p_i / q_i}, \quad \sum_i |c_i| \leq \sqrt{\sum_i v_i} \sqrt{\sum_i p_i / q_i} < \infty,$$

since $q_i \geq 1/3$. Write $X_i = c_i + Y_i$. With probability one only finitely many Y_i are nonzero, and every finite configuration of positive-probability states is an atom of the full law, because $\prod_i q_i > 0$.

Two compactness observations retain the endpoint conventions needed below. For fixed M , the set of sums using at most M nonzero Y_i is compact. In a sequence of such sums, each of its at most M selected indices can be made either constant or divergent; divergent indices contribute zero in the limit. Every limit is another finite configuration. If all the selected nonzero values are positive, then, for each $b > 0$, these sums avoid an interval $[b - \eta, b)$ for some $\eta > 0$. Otherwise a sequence approaching b from below has fixed terms whose sum is b and escaping positive terms, so it is eventually at least b , a contradiction.

If the maximizing threshold z is itself an atom of mass $m > 0$, all sufficiently late gaps from the mode to another state are negative. Indeed, for a positive gap $d_i = u_i - c_i \rightarrow 0$, the modal-branch CDF $H_i(w) = \mathbb{P}(S - X_i + c_i \leq w)$ satisfies $\|H_i - F\|_\infty \leq p_i \rightarrow 0$. Hence

$$H_i(z - d_i) - H_i(z) \rightarrow -m.$$

The stationarity equation (3.4) instead makes this difference tend to zero, since both states and their gap tend to zero. This excludes such a sequence.

Fix K so that $q_j \geq 3/4$ for $j \geq K$, and select a nonmodal state of an arbitrarily late nonconstant coordinate $i \geq K$. Set $d_i = u_i - c_i$ and $S_i^0 = S - X_i + c_i$. Its exact mass-transfer derivative is

$$g_i = -\text{sgn}(d_i)\mathbb{P}(E_i), \quad E_i = \begin{cases} \{z - d_i < S_i^0 \leq z\}, & d_i > 0, \\ \{z < S_i^0 \leq z - d_i\}, & d_i < 0. \end{cases} \quad (6.1)$$

Let N_i count the nonmodal outcomes among $j \geq K$, $j \neq i$. For every fixed integer M ,

$$E_i \subset \{N_i > M\} \quad \text{for all sufficiently late } i. \quad (6.2)$$

If z is not an atom, it has positive distance from the compact set of all configurations with at most $M + K$ nonmodal states. The intervals in (6.1) shrink to z , proving (6.2). If z is an atom, take K beyond the positive-gap exclusion. Conditional on any of the finitely many early prefix values, write $S_i^0 = b - \sum_{j \geq K, j \neq i} Z_j$ with $Z_j \geq 0$. Its pivotal interval is exactly

$$\sum_{j \geq K, j \neq i} Z_j \in [b - z - |d_i|, b - z).$$

For $b - z \leq 0$ it is empty; otherwise the one-sided compactness observation proves (6.2). The right endpoint is excluded, as required.

Vary every nonmodal probability for $j \geq K$, $j \neq i$, by the factor $1 + t$, taking the compensating mass from the mode. Both signs of $|t| < 1/2$ are admissible. The original product-law score is

$$L_i = \sum_{j \geq K, j \neq i} \ell_j, \quad \ell_j = \begin{cases} 1, & Y_j \neq 0, \\ -p_j/q_j, & Y_j = 0. \end{cases} \quad L_i \geq N_i - \Lambda, \quad \Lambda = \sum_{j \geq K} p_j/q_j < \infty.$$

This infinite product variation is twice differentiable for every bounded observable. Its density is

$$(1 + t)^{N_i} \prod_{Y_j=0} (1 - tp_j/q_j).$$

The product and its first two derivatives are dominated, uniformly in i , by a constant times a polynomial in N_i times $(3/2)^{N_i}$. These bounds are integrable because $\mathbb{E}b^{N_i} \leq \exp((b-1)\sum_{j \geq K} p_j)$ for every $b \geq 1$. Differentiating (6.1) in this direction gives the exact mixed raw-CDF derivative

$$g_{i,t} = -\text{sgn}(d_i)\mathbb{E}[\mathbf{1}_{E_i} L_i].$$

By (6.2), $\mathbb{E}[L_i | E_i] \rightarrow \infty$ whenever these events have positive probability.

Let $h_i = \max\{|x| : x \in \text{supp } X_i\} \rightarrow 0$. The mass-transfer stationarity equation and

$$|u_i^2 - c_i^2| \leq 2h_i|d_i|, \quad |r'_i| \leq 6h_i^2|d_i|$$

give

$$g_i/d_i \rightarrow -\phi(z), \quad \mathbb{P}(E_i)/|d_i| \rightarrow \phi(z) > 0. \quad (6.3)$$

It follows that $|g_{i,t}|/|d_i| \rightarrow \infty$.

On the other hand, form (3.1) with two parameters: the selected mass transfer s in coordinate i and the collective variation t just defined. Uniformly in late i , its Hessian satisfies

$$|J_{ss}| \leq K_1 d_i^2, \quad |J_{st} - g_{i,t}| \leq K_2 |d_i|, \quad |J_{tt}| \leq K_3. \quad (6.4)$$

To check these bounds, the selected derivatives are $m_s = d_i$, $B_s = u_i^2 - c_i^2$, $B_{ss} = -2d_i^2$, $F_{ss} = 0$, $F_s = O(|d_i|)$ and $|R_{ss}| \leq 18h_id_i^2$ by (3.4). All collective moment derivatives through order two are bounded by summability of p_j and the common support bound. The raw CDF derivatives are uniformly bounded by the product estimate above. Mixed moment derivatives vanish because the two parameters affect disjoint coordinates. The normal normalization is smooth near $(B, m) = (1, 0)$, proving (6.4). Both parameters are two-sided, so negative semidefiniteness gives $|J_{st}|^2 \leq K_1 K_3 d_i^2$. This contradicts the divergent mixed CDF ratio. Therefore an infinite maximizing sum is nonatomic.

Conversely, if $\sum_i (1 - q_i) < \infty$, the same modal-drift argument and Borel–Cantelli show that the series is a countable discrete law. This proves the modal-deficit assertion and completes the proof of Theorem 1.2.

7 Support motions and the completion rank

We prove Theorem 1.3 directly at the maximizing representation. Suppose, to the contrary, that finite sets of arbitrarily late coordinates admit nonconstant state functions with constant sum on their completion sets. Subtract their individual means, absorbing the sum into the constant c . Write

$$\mathbb{E}f_i(X_i) = 0, \quad w_i = \mathbb{E}f_i(X_i)^2, \quad W = \sum_{i \in I} w_i > 0, \quad V = \sum_{i \in I} v_i, \quad h = \max_{i \in I} \max |\text{supp } X_i|.$$

Along the supposed sequence, $h + V \rightarrow 0$.

Move each raw state x_i to $x_i + tf_i(x_i)$ and move the raw threshold to $z + tc$. Every completion pattern has the same residual threshold as before. For each other pattern the residual threshold lies outside the closed support of T_I , in an open interval where its CDF is constant. There are finitely many patterns, so one two-sided interval of sufficiently small t works for all of them. Distinct coordinate states remain distinct after reducing the interval if necessary. Thus the raw CDF is exactly unchanged, also when a completion is an atom or the endpoint of a gap. This reasoning permits noncompact remainder support.

Choose j maximizing w_j/v_j among the nonconstant functions, and put

$$\begin{aligned} X &= X_j, \quad f = f_j(X_j), \quad v = v_j, \quad w = w_j, \\ m &= \mathbb{E}Xf, \quad a_1 = \mathbb{E}X^2f, \quad b_1 = \mathbb{E}Xf^2, \quad M = \sum_{i \in I} \mathbb{E}[X_i f_i(X_i)]. \end{aligned}$$

Then

$$v/w \leq V/W. \tag{7.1}$$

Tilt the original law of coordinate j to $(1 + sf)d\text{Law}(X)$, and apply the support motion. Both signs of s are admissible because its support is finite. Its moved mean is $sm + stw$. The centered threshold x and total variance B have derivatives

$$\begin{aligned} x_s &= -m, \quad x_t = c, \quad x_{st} = -w, \quad B_s = a_1, \quad B_{ss} = -2m^2, \\ B_t &= 2M, \quad B_{tt} = 2W, \quad B_{st} = 2b_1. \end{aligned} \tag{7.2}$$

The exact raw CDF satisfies $F_{ss} = F_t = F_{st} = F_{tt} = 0$. Cauchy–Schwarz gives

$$|m| \leq \sqrt{vw}, \quad |a_1| \leq h\sqrt{vw}, \quad |b_1| \leq hw, \quad |M| \leq \sqrt{VW}.$$

For clarity, the separately recentered third moments have the following derivatives; all expectations involving X, f concern the selected coordinate:

$$\begin{aligned}
R_t &= 3 \sum_{i \in I} \mathbb{E}[X_i | X_i | f_i], & R_{tt} &= 6 \sum_{i \in I} \mathbb{E}[|X_i| f_i^2], \\
R_s &= \mathbb{E}[|X|^3 f] - 3m \mathbb{E}[X | X|], & R_{ss} &= 6m^2 \mathbb{E}[X] - 6m \mathbb{E}[X | X | f], \\
R_{st} &= 3 \mathbb{E}[X | X | f^2] - 3w \mathbb{E}[X | X|] - 6m \mathbb{E}[|X| f]. & & (7.3)
\end{aligned}$$

It follows that

$$\begin{aligned}
|R_t| &\leq 3h\sqrt{VW}, & |R_{tt}| &\leq 6hW, \\
|R_s| &\leq 4h^2\sqrt{vw}, & |R_{ss}| &\leq 12hvw, & |R_{st}| &\leq 12h^2w.
\end{aligned}$$

These bounds use $\mathbb{E}X^4 \leq h^2v$ and do not require a lower bound on any state probability.

First stationarity in (3.1) gives

$$\phi c = 2AM - CR_t, \quad F_s = -\phi m - Aa_1 + CR_s.$$

Thus $|c| \leq K\sqrt{VW}$ and $|F_s| \leq K\sqrt{vw}$. Using (7.2)–(7.3) in its Hessian now gives

$$|J_{ss}| \leq K_1vw, \quad |J_{tt}| \leq K_2W, \quad J_{st} = \phi w + O((h+V)w).$$

Indeed the term $-\phi x_{st}$ is ϕw , direct variance and third-moment terms are $O(hw)$, and products of first derivatives are $O(\sqrt{VWvw}) = O(Vw)$ by (7.1). All constants depend only on the fixed maximizing representation and threshold. Negative semidefiniteness would imply, for sufficiently late sets,

$$\phi^2 w^2 / 4 \leq K_1 K_2 vwW \leq K_1 K_2 Vw^2,$$

contradicting $V \rightarrow 0$. This proves Theorem 1.3.

Proof of Corollary 1.4. Fix a late cut and absorb all earlier blocks into the finite prefix; let it have N distinct sum values. Select $M > 2N$ consecutive remaining blocks. The unchanged infinite future has support in an interval of positive width $W = \sum_{l \text{ after these blocks}} H_l$. Distinct vectors of selected block-sum values have total sums at least W apart: at their first differing block k , their difference in absolute value is at least

$$\delta_k - \sum_{l > k, l \text{ selected}} H_l \geq W.$$

For each prefix atom, at most two such vectors put the maximizing threshold inside the translated closed future interval. Thus at most $2N$ selected block vectors can complete through the actual remainder support, including endpoints.

There are $M+1$ unknowns $a_1, \dots, a_M, c$ and at most $2N$ homogeneous equations $\sum_k a_k u_k = c$ on these vectors. A solution exists with some $a_k \neq 0$. Apply Theorem 1.3 to the individual functions $f_i(x) = a_k x$ for coordinates in block k . At least one is nonconstant because each block is nonconstant. Their sum is constant on every completion pattern, a contradiction for a sufficiently late cut. The total variance and largest coordinate radius of the selected finite family tend to zero, independent of M . $\square$

A Rational checks for the Gaussian Fourier bound

The finite calculation in Lemma 5.3 uses only rational arithmetic. For $x \geq 0$ define

$$E_n(x) = \sum_{k=0}^n \frac{x^k}{k!}, \quad U_n(x) = E_n(x) + \frac{x^{n+1}}{(n+1)!} \frac{1}{1 - x/(n+2)} \quad (0 \leq x < n+2).$$

Then $E_n(x) \leq e^x \leq U_n(x)$, by the positive exponential series and a geometric bound on its remaining terms. The exponent $-t^2/2 + (269/5000)t^3$ is decreasing on $[0, 6]$. With $d = 3/100$, $l_j = jd$, $r_j = (j+1)d$, the exact upper sum

$$\sum_{j=0}^{199} \frac{d r_j^2}{E_{40}(l_j^2/2 - (269/5000)l_j^3)} < \frac{23}{10} \quad (\text{A.1})$$

therefore bounds I . Integration by parts and the normal Mills bound give

$$J \leq \frac{6025}{216} e^{-108/25} \leq \frac{6025}{216 E_{40}(108/25)} < \frac{3}{8}.$$

The remaining rational comparisons are

$$\begin{aligned} \frac{U_{24}(25/54)}{6(157/50)} &< \frac{17}{200}, & \frac{11}{19E_{40}(361/289)} &< \frac{1}{6}, \\ \frac{25}{61} + \frac{(3051/5000)(17/20)2}{(443/250)(9/10)(237/250)} &< \frac{11}{10}. \end{aligned}$$

The last expression bounds the Gaussian-convolution density because

$$(61/25)^2 < (19/10)(157/50), \quad (443/250)^2 < 157/50, \quad (237/250)^2 < 9/10.$$

The normal tail bound follows from $T > 7$, $\beta < 3/10$, $\pi > 3$, $c > 1$, $e > 5/2$, and $(5/2)^{24} > 10^8$. All these are finite rational inequalities, including (A.1); the executable companion `gaussian_window_certificate.py` evaluates them using Python's standard-library rational type and asserts each comparison. It contains no floating-point quadrature or external numerical library. Its final exact comparison is

$$\frac{1182125003}{3000000000} < \frac{2}{5}.$$

The classical numerical constant 0.3051 in (5.7) is an external theorem of Korolev and Shevtsova; this finite check verifies the additional comparisons used here.

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